^{9}Be(^{37}Ca,X) 2024Dr01

Isotope discovery of ^{34}K.

2024Dr01: a 72-MeV/nucleon ^{37}Ca secondary beam was produced via the fragmentation of a 140-MeV/nucleon $^{40}Ca^{20+}$ primary beam impinging on a 0.5-mm-thick Be target. Experimental setup includes the CAESium-iodide scintillator ARray (CAESAR) for detecting γ rays, a DSSD-CsI(Tl) ΔE-E Ring Telescope for detecting protons, a Scintillating-Fiber Array (SFA) and the S800 spectrograph for detecting heavy residuals. Measured E_p, Eγ, and E(residual). Deduced levels, mass excess from total decay-energy spectra using the invariant-mass method. Comparisons with shell-model calculations with the USDC Hamiltonian.

^{34}K Levels

E(level)†	J$^\pi$‡	Comments
0	(1$^+$)	Total decay energy E_T=608 *17*. Mass excess=−1487 keV *17*, deduced by 2024Dr01 from their measured total decay energy and mass of ^{33}Ar from 2021Wa16.
401 *25*	(2$^+$)	E(level): from total decay energy E_T=1009 *18*.
≈1240		E(level): from total decay energy E_T≈=1850.
≈1810		E(level): from total decay energy E_T≈=2420.

† From the measured total decay−energy spectrum of p+^{33}Ar.

‡ Assigned by 2024Dr01 based on shell model. Parentheses are added by evaluators.